HEIN HTET

Edge ML Engineer | On-Device AI & Embedded Systems

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PROFESSIONAL SUMMARY

Engineer with 4+ years of hands-on experience taking ML models from training to real-time execution on constrained hardware: neural control policies on NVIDIA Jetson, voice agents for ESP32 devices, and a SLAM pipeline with its numerical libraries ported to dependency-free C. Combines ML (PyTorch, ONNX Runtime, reinforcement learning) with systems engineering in C, C++ and Rust: real-time control loops, CAN bus, embedded firmware and edge-cloud agent architecture. Currently AI Architecture Lead at Magick Mind, building the agent SDK and voice runtime behind its devices and robots.

TECHNICAL SKILLS

- **Edge ML & Inference:** ONNX Runtime, TorchScript, PyTorch, Ollama (local LLM serving), real-time model deployment, edge/cloud build targets
- **Languages:** C, C++, Rust, Python, Go
- **Embedded & Real-Time:** NVIDIA Jetson Orin Nano, ESP32, STM32, CAN bus (SocketCAN), I2C, real-time multithreading, lock-free concurrency, Linux, ROS 2
- **Robotics & Perception:** Reinforcement learning (PPO), Isaac Lab, sim-to-real transfer, MuJoCo, LiDAR-inertial odometry (FAST-LIO), visual SLAM (ORB-SLAM2), IMU integration
- **Numerical Computing:** Linear algebra and optimization from scratch (LU, Cholesky, SVD, Levenberg-Marquardt)
- **AI Agents & Voice:** LLM agents, tool calling, real-time voice pipelines (VAD, STT, LLM, TTS, barge-in), Gemini Live, OpenAI Realtime
- **Platform:** Tokio, gRPC, Zenoh, Docker, Kubernetes, CI/CD (GitHub Actions)

WORK EXPERIENCE

Magick Mind | Bangkok, Thailand | Apr 2026 - Present

AI Architecture Lead & Team Co-Lead

AI agent infrastructure for developers: memory, persona and reasoning services with Python and Rust SDKs, working with any LLM provider, plus a companion app and ESP32 companion robots.

- **Architecture & leadership:** Lead AI architecture and co-lead a **~10-engineer team** across a **17-service platform** (Go, Python, Rust) spanning agents, memory, inference and voice.
- **Real-time voice for devices:** Led development of the Rust voice server: Gemini Live plus a cascaded VAD, STT, LLM and TTS pipeline with barge-in, language hints and noise gating, designed for up to **50 concurrent sessions** and **~500 ms time-to-first-audio**. LLM tool calls over the live stream drive **2 ESP32 companion robots**' motion and gestures, with safety stops.
- **Edge agent SDK:** Contributed Mindroid, the company's open-source Rust agent SDK, with feature-gated builds for cloud or edge targets and a gated skill system; built its tool-calling engine and realtime voice layer (Gemini Live and OpenAI Realtime).
- **Embodied agents:** Built an LLM-to-robot adapter that exposes robot capabilities as agent tools over gRPC and Zenoh, validated on **3 MuJoCo robots** (differential drive, Unitree Go2, Stretch) and inferenced on an SO-ARM101 arm.
- **Device security:** Created the device backend (argon2 auth, rotating refresh tokens, robot pairing) and designed its token broker, which gives each device scoped, short-lived credentials so tenant API keys never reach the hardware.
- **Latency-aware inference:** Designed the fast/smart inference path: a low-latency fast path for real-time interaction and a smart path that runs several frontier models in parallel and aggregates their answers.

Magick Robotics | Bangkok, Thailand | Mar 2025 - Mar 2026

Project Lead & Senior Robotics Engineer

- **Leadership:** Led a small cross-disciplinary team (mechanical, hardware, ML) building the Tet Hlan **12-DOF** bipedal humanoid; owned the software architecture, roadmap and milestones.
- **RL training:** Led RL training in Isaac Lab with PPO across **4096 parallel environments** (**44-dim** observation, 12 joint-position actions, **50 Hz** policy over **200 Hz** physics) with domain randomization; personally designed and tuned the reward terms.
- **On-device inference:** Deployed the policy on NVIDIA Jetson Orin Nano at **50 Hz** through TorchScript and ONNX Runtime, taking it from walking in simulation to stable standing balance on hardware.

- **Real-time control:** Wrote the real-time C++ motor-control stack: **12 CubeMars motors** over **2 x 1 Mbit/s CAN** buses, one **1 kHz** thread per bus reading lock-free atomic targets, Ruckig-smoothed commands and parallel-linkage ankle kinematics from a **0.5-degree** lookup table.
- **Sensing:** Wrote the **200 Hz** BNO055 IMU node over I2C and integrated a Livox MID-360 LiDAR with FAST-LIO odometry; migrated the platform to ROS 2 Humble.

Panasonic Research Lab | Singapore | Oct 2023 - Jul 2024
Student Researcher

- Rewrote ORB-SLAM2 for constrained compute: implemented the numerical and optimization libraries (LU, Cholesky, SVD, Levenberg-Marquardt) from scratch and ported the **6-module** visual localization pipeline to dependency-free C.

KMUTT Deep Learning Lab | Bangkok, Thailand | Aug 2022 - Jun 2025
Research Assistant

- Built a speech conversion model in PyTorch (kNN-VC, WavLM, HiFi-GAN) that **outperformed MaskCycleGAN-VC and SELM baselines** (**Gold Prize**, Student Innovation Challenge Thailand 2024) and shipped it as a mobile app with a cloud backend and end-to-end inference **under 3 seconds**.

Singapore University of Technology and Design (SUTD) | Singapore | Jun 2024 - Aug 2024
Student Researcher

- Benchmarked **4 open-weight LLMs** (Llama 3, Phi-3, Mistral, Gemma) served locally with Ollama, using **3 custom evaluation metrics** alongside ROUGE, BLEU and F1.

Mu Space Corp | Bangkok, Thailand | Feb 2022 - Jul 2022
Embedded Engineering Intern

- Developed satellite firmware across **5 STM32 architectures**, including an ion thruster control subsystem.

EDUCATION

King Mongkut's University of Technology Thonburi (KMUTT) | Bangkok, Thailand | Jul 2025
B.Eng. Electronic and Infocommunication Engineering | GPA 3.71/4.00, First Class Honours

AWARDS & LEADERSHIP

- **Gold Prize**, Student Innovation Challenge Thailand 2024
- Founder, Clash of Robots, Myanmar's first national robotics competition (2018 - 2020)
- English: TOEIC 970/990